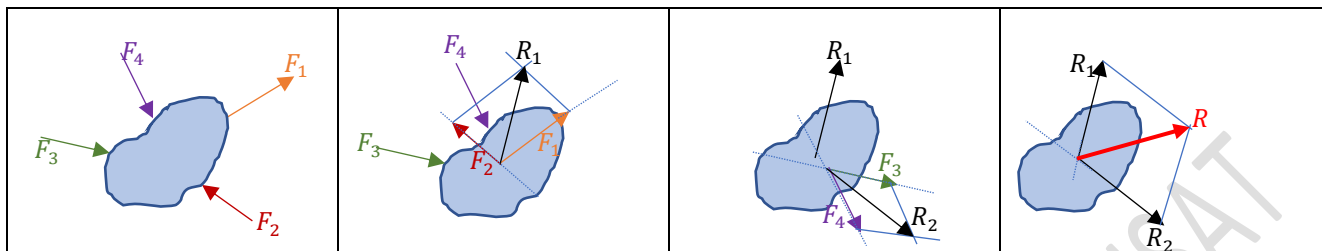


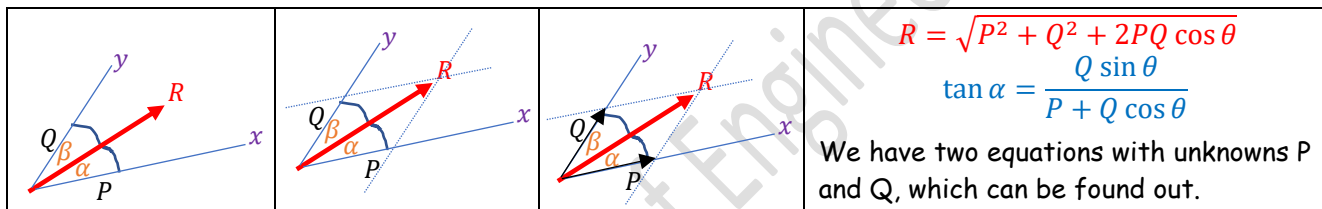
Composition and Resolution of forces

The process of finding one resultant force from a system of forces is known as composition of forces. We have seen that the resultant of two forces can be found out by using parallelogram law. When there are a greater number of forces, the parallelogram law is applied to two forces successively till we get a single resultant force R .



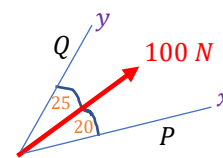
The forces F_1 and F_2 are transmitted along their lines of action to make them meet at one point, their resultant R_1 is found out using parallelogram law. Then F_3 and F_4 are transmitted along their lines to their point of concurrency and their resultant R_2 is found out. Then the resultant R can be found out from R_1 and R_2 .

The process of finding out the components of a force is known as resolution. Again, we can use the parallelogram law to do the resolution. Suppose, a force R is to be resolved into components P and Q along directions x and y making angles α and β respectively.



Example 1

Resolve a force 100 N into its components along the directions shown



$$R = 100 \text{ N} \quad \ll \gg \quad \alpha = 20^\circ \quad \ll \gg \quad \theta = 45^\circ$$

$$100 = \sqrt{P^2 + Q^2 + 2PQ \cos 45} \quad \gg \gg \quad 100^2 = P^2 + Q^2 + 2PQ \cos 45$$

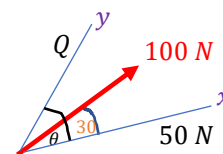
$$\tan 20 = \frac{Q \sin 45}{P + Q \cos 45} \quad \gg \gg \gg \quad 0.364P + 0.257Q = 0.707Q \quad \gg \gg \gg \quad P = 1.24Q$$

$$100^2 = (1.24Q)^2 + Q^2 + 2(1.24Q)Q \cos 45 \quad \gg \gg \gg \quad Q = 48.3 \text{ N}$$

$$P = 59.8 \text{ N}$$

Example 2

If one component of a force 100 N is 50 N at 30° , what is the magnitude and direction of the other?



$$R = 100 \text{ N} \quad \ll \gg \gg \gg \quad P = 50 \text{ N} \quad \ll \gg \gg \gg \quad \alpha = 30^\circ \quad \ll \gg \gg \gg \quad \theta = ? \quad \ll \gg \gg \gg \quad Q = ?$$

$$100 = \sqrt{50^2 + Q^2 + 2 \times 50 \times Q \cos \theta} \quad \gg \gg \gg \quad 100^2 = 50^2 + Q^2 + 100Q \cos \theta \quad \gg \gg \gg \quad 100Q \cos \theta = 7500 - Q^2$$

$$\tan 30 = \frac{Q \sin \theta}{50 + Q \cos \theta} \quad \gg \gg \gg \quad 28.9 + 0.577Q \cos \theta = Q \sin \theta \quad \gg \gg \gg \quad Q \sin \theta = 28.9 + 0.577 \left(\frac{7500 - Q^2}{100} \right) \quad \gg \gg \gg \quad 100Q \sin \theta = 7216 - 0.577Q^2$$

Squaring and adding

$$100^2 Q^2 = 7500^2 - 15000Q^2 + Q^4 + 7216^2 - 8333Q^2 + 0.333Q^4$$

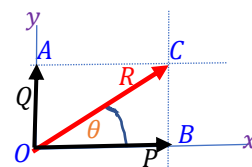
$$1.333Q^4 - 33333Q^2 + 108320656 = 0 \quad \gg \gg \gg \quad Q^2 = 3839, 21167 \quad \gg \gg \gg \quad Q = 62 \text{ N}, 145 \text{ N}$$

$$\theta = 53.9^\circ, 158.9^\circ$$

If a force is resolved into its components along two mutually perpendicular directions, they are called **rectangular components**.

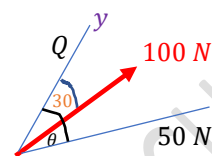
From the $\triangle OBC$, $\cos \theta = \frac{OB}{OC} = \frac{P}{R} \gggg P = R \cos \theta$

$$\sin \theta = \frac{CB}{OC} = \frac{Q}{R} \gggg Q = R \sin \theta$$



Exercise 1

If one component of a force $R = 100\text{ N}$ is $P = 50\text{ N}$ and the other component Q should be at 30° counter clockwise from R , what are the magnitude of Q and direction of P ?

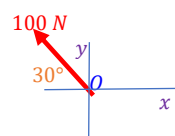
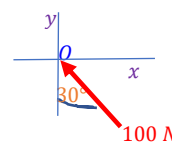
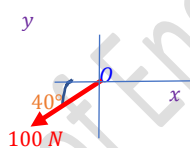
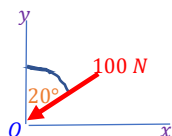
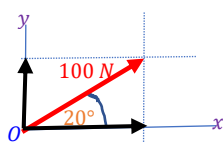


Exercise 2

If a force $R = 100\text{ N}$ is to be split into two forces $P = 55\text{ N}$ and $Q = 55\text{ N}$ on either side of it, what are the angles they make with R ?

Exercise 3

Resolve the given force into its rectangular components



Reference

1. Engineering Mechanics: Timoshenko and Young, McGraw Hill
2. Mechanics for Engineers: Beer & Johnston, McGraw Hill
3. Engineering Mechanics: Merriam & Kraig, Wiley
4. Engineering Mechanics: Hibbeler, Pearson

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